

Hard and Soft Decision Decoding Using Viterbi Algorithm

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Abstract— Any information that has to be transmitted undergoes a number of processes including the noise channels before it is extracted by the receiver. This brings in the need for reliable reception of information with minimum error rate. The main aim of this paper is to shed light on Viterbi algorithm as an error detection and correction method used to maintain the reliability of the system. This paper focuses on Viterbi Decoder, an efficient Hard/Soft algorithm devised to estimate the most likely path for convolutionally encoded data with implementation details on Hard and Soft decision techniques through various communication channels.

Keywords – Viterbi Algorithm, communication channel, hard decision, soft decision.

I. INTRODUCTION

Wireless connectivity is now becoming a fast developing reality for both voice and data. Data transmission and reception over wireless channels can be greatly affected by distortion, interference and noise. This causes errors to be induced in the received signal. This brings in the need for correction of induced error to improve the efficiency of the communication system. Error correction has therefore become an integral part of data communication. An information message goes through a certain fixed number of blocks for transmission and reception as a part of communication system[2]. The block diagram of a communication system is as shown in Fig. 1.

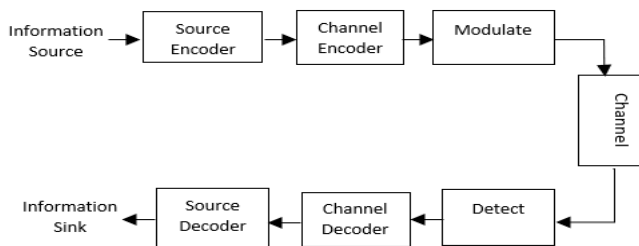


Fig. 1. Communication System

II. CONVOLUTIONAL ENCODER

Any information/message, before being transmitted has to be encoded first. Channel encoding is done so as to add redundant data so as to improve performance of error. The encoding results in the formation of a code word which would then be transmitted. One way of encoding data is through Convolutional Encoding. In convolutional encoding, the message is transmitted through certain number of shift registers. The contents of shift register are multiplied by respective term in generator matrix and are then added together to generate respective code words.

The generator polynomials are given by the following equations:

$$g1(x) = 1 + x + x^2 \quad (1)$$

$$g2(x) = 1 + x^2 \quad (2)$$

Convolutional encoders are described by using the following parameters:

k = Number of input symbols

n = Number of output signals

K = Constraint length

m = Number of memory registers

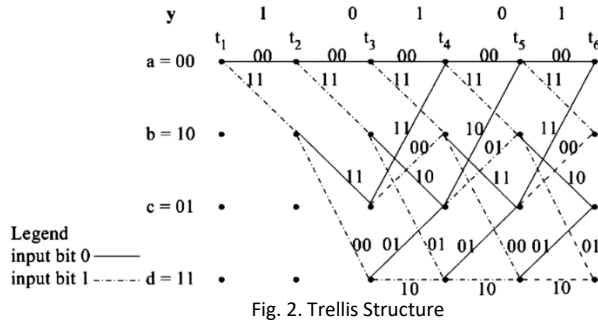
Code rate is measured as bandwidth efficiency of the code. Code rates can be given by $\frac{1}{2}$, $\frac{1}{4}$ and so on. It is defined by

$$r = \frac{k}{n} \quad (3)$$

The bits are then passed through the adders and then the output from the adders are sampled. The sampled result gives rise to the code symbols for transmission.

The encoding can be represented in the form of a tree like structure which is referred to as Trellis diagram[10]. Trellis Structures are structures that are used for decoding codes in which each node representing a state at a given point of time.

Trellis diagram is used to decipher the possible patterns in which data is received. An example of a trellis diagram is as shown in Fig. 2.



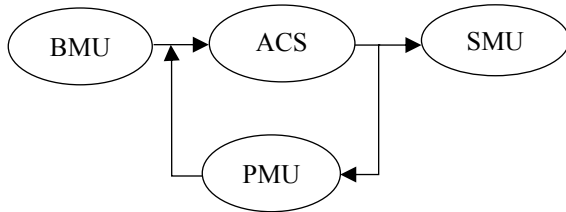
III. VITERBI DECODER

Viterbi Algorithm was first developed by Andrew J. Viterbi. Viterbi Decoder is used to decode the convolutionally coded data. This convolutionally coded data is prone to errors after passing through a communication channel[1]. Viterbi Decoder is used as a forward error correction technique to eliminate errors induced in communication channel[5].

Viterbi Algorithm uses Add, Compare and Select (ACS) method to determine the most likely path sequence.

Decoding is said to be hard decision if the signal is quantized in binary using just two bits, i.e., 0 or 1 while it is said to be soft decision if quantization is done on multiple levels[3].

The block diagram of Viterbi Algorithm is as shown in Fig. 3.



1. Branch Metric Unit :

A branch metric unit is responsible for calculating the hamming distance for a branch in hard decision Viterbi decoding and Euclidian distance for a branch in soft decision Viterbi decoding.

Hamming distance is obtained by ex-oring the received symbol as well as actual code symbol and counting the number of 1's.

2. Path Metric Unit :

A path metric unit is computed by the modulo 2 adders. It is responsible for comparing and selecting the appropriate branch.

The appropriate branch is given by the value for which hamming distance/Euclidian distance is lower. This particular value is computed for each path and stored as a buffer for each node.

3. Trace Back Unit/ Survivor Unit :

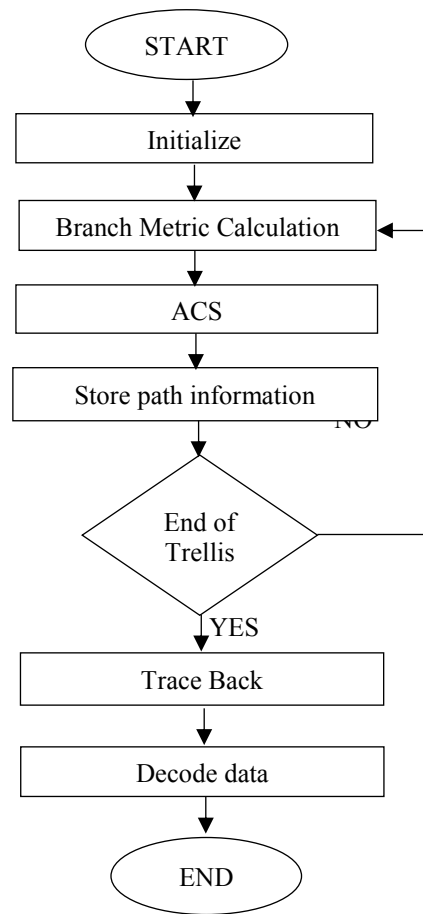
The decoded sequence is the result of trace back. Trace back is performed from the final state to the beginning state.

The optimal path is given by the maximum likelihood path. This information is stored and then traced back path is obtained.

4. Add, Compare, Select :

This unit comprises of comparing the branch metrics and then selecting the branch with lower hamming/Euclidian distance while the other branch is deleted away.

Fig. 4 depicts the flowchart of Viterbi method ,i.e., the steps to be followed for the Viterbi algorithm.



The functionality remains the same for both hard and soft decision decoder, where the only difference is that hard decision computes the Hamming distance while soft decision computes the Euclidian distance as the branch metric[7].

In this way Viterbi Decoder functions to mitigate the errors introduced as an effect of a noisy communication channel.

IV. ALGORITHM

Iterative methodology has been used to improvise on the accuracy of the infrared sensor.

The algorithm for Viterbi Decoder defines the steps taken for the accomplishment of maximum likelihood path to reduce the bit error rate in bit transmission and reception in a communication channel[6].

The algorithm for the implementation of the Viterbi Decoder is as follows:

Step 1: The convolutionally encoded data is received from the communication channel with induction of noise element.

Step 2: The branch metric is calculated for each branch of the received data which is in the form of a Trellis Structure.

Step 3: The branch metrics of for every node or point in the Trellis structure are compared and the branch with the lowest metric is saved while all the other branches are deleted. This is called as Add, Compare and Select operation.

Step 4: Every Add, Compare Select operation is followed by checking for condition if the end of Trellis has been reached. If no, then the ACS method is operated for the next node.

Step 5: When the end of the Trellis is reached, then traceback algorithm starts off, which traces back from the end node to the beginning node whilst selecting the most optimal path.

Step 6: The path is traced out but the data however, is still encoded. Therefore, the received data is decoded to obtain the data with minimum bit error rate.

V. IMPLEMENTATION

Viterbi Decoder is implemented on MATLAB Simulink through convolutional encoder followed by passage of AWGN channel and reception followed by Viterbi decoder[4].

The input for the system is given as Bernoulli Binary Generator which generates random binary bits i.e., in the form of 0's and 1's. The digital stream of bits are then passed through a block for Convolutional Encoder. The convolutionally encoded data can then be modulated using any of the digital modulation techniques. This step is not mandatory and can be skipped as well. The data is modulated generally for efficient transmission and reception of data bits. Here the signal to be transmitted is modulated using the Binary Phase Shift Keying Modulator[8]. This data is then passed through Additive White Gaussian Noise (AWGN) Channel where noise gets introduced.

The data is extracted from the channel is then demodulated using Binary Phase Shift Keying Demodulator.

The demodulated data is then passed through a Viterbi Decoder block which executes Viterbi Algorithm for decoding the transmitted data and minimizing the bit error rate.

The MATLAB Simulink model for the implementation of hard decision decoder when passed through an AWGN channel is as shown in Fig. 5.

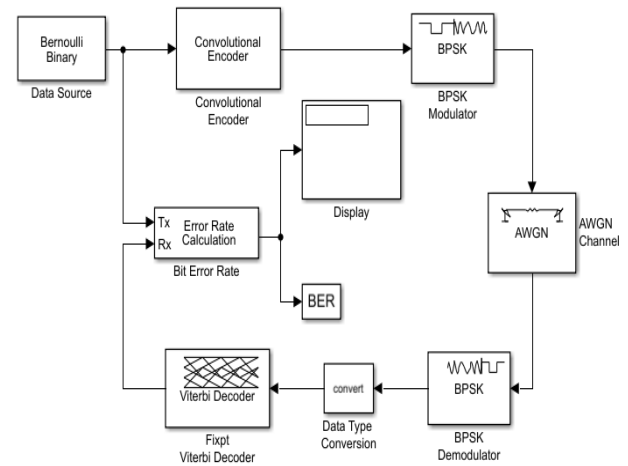


Fig. 5. Hard Decision Viterbi Decoder through AWGN channel

The MATLAB model for the implementation of soft decision decoder when passed through an AWGN channel is as shown in Fig. 6.

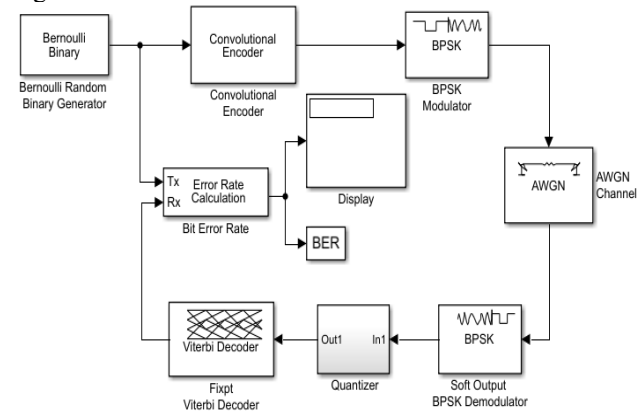


Fig. 6. Soft Decision Viterbi Decoder through AWGN channel

Fig. 7 shows the MATALB Simulink block for hard decision decoder through Rayleigh Channel.

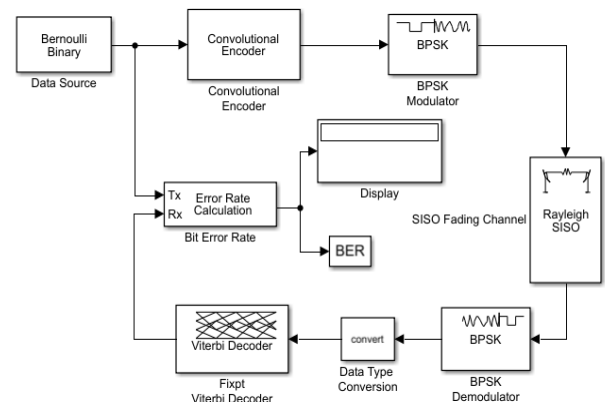


Fig. 7. Hard Decision Viterbi Decoder through Rayleigh channel

Rayleigh channel is set on MATLAB Simulink by SISO fading and the maximum Doppler shift is set to 0.001 Hz with the initial seed being 64.

Fig. 8 shows the MATABL Simulink block for soft decision decoder through Rayleigh Channel.

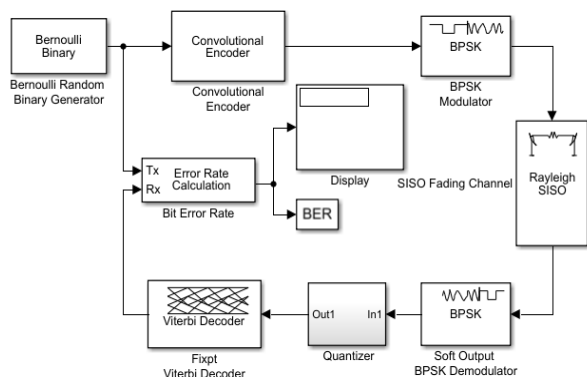


Fig. 8. Soft Decision Viterbi Decoder through Rayleigh Channel

The MATLAB Simulink model for the implementation of hard decision decoder through an Rician channel is shown in Fig. 9.

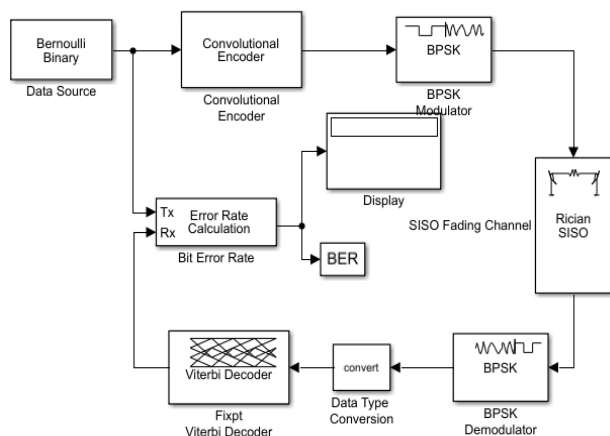


Fig. 9. Hard Decision Viterbi Decoder through Rician Channel

The MATLAB Simulink model for the implementation of soft decision decoder through an Rician channel is shown in Fig. 10.

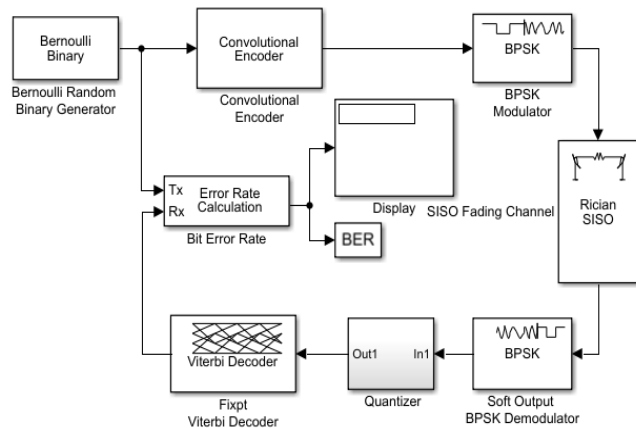


Fig. 10. Soft Decision Viterbi Decoder through Rician channel

The Error Rate Calculation block shows three readings, which are error rate, total number of errors and total number of comparisons respectively.

VI. RESULTS

The MATLAB Simulink models for hard and soft decision decoder when passed through AWGN, Rayleigh and Rician fading are simulated.

It can be observed that the performance of soft decision decoder is better than a hard decision decoder and therefore provides a lower Bit Error Rate (BER)[9].

The results of the simulations from the models are as shown in TABLE I.

TABLE I.

Type of Decoder	Communication Channel	Bit Error Rate (BER)
Hard Decision Viterbi	AWGN	$2.027 * 10^{-3}$
	Rayleigh	$3.219 * 10^{-3}$
	Rician	$5.29 * 10^{-3}$
Soft Decision Viterbi	AWGN	$1.136 * 10^{-4}$
	Rayleigh	$3.044 * 10^{-3}$
	Rician	$7.335 * 10^{-4}$

It can be seen that in all of the three communication channels, i.e., AWGN, Rayleigh and Rician channel, the performance is better in Soft decision decoding technique when compared to hard decision decoding technique. This is because of quantization done on multiple levels in soft decision as opposed to just binary digits in hard decision Viterbi algorithm.

The BER is at the scale of 10^{-3} for Hard decision through all three communication channels while it is at the rate of 10^{-4} in Soft decision for AWGN and Rician fading channels with the exception of Rayleigh channel where it is to the scale of 10^{-3} . Though while Soft Decision gives near to accurate results, the realization of Soft Decision is much more complex in comparison to Hard Decision which is conveniently easier.

VII. CONCLUSION

In this paper, we have presented the theoretical background for Hard and Soft Decision Viterbi Decoding followed by the implementation steps for realizing the application through AWGN, Rayleigh and Rician fading channels on MATLAB Simulink and thereby observed the performance of Viterbi Decoding algorithm through various communication channels on account of BER. It is observed that for any communication channel like AWGN, Rayleigh or Rician fading; the performance is much better in Soft decision decoding in contrast to hard decision decoding owing to multiple levels of quantization in Soft decision technique.

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REFERENCES

- [1] Akkidas, Davidrichesemmanuel. (2015). VITERBI ALGORITHM. Journal of Experimental Algorithmics.
- [2] A.J. V. Omura, "*Principles of Digital Communication and Coding*", 1979.
- [3] D. H. Meliani and A. Guellal, "*Comparison between Viterbi Algorithm Soft and Hard Decision Decoding*", pp 9.
- [4] J. He, Z. Wang, Z. Cui and L. Li, "Towards an optimal trade-off of Viterbi Decoder Design," *2009 IEEE International Symposium on Circuits and Systems*, Taipei, 2009, pp. 3030-3033.
- [5] Mr. Sandesh Y.M et al *Int. Journal of Engineering Research and Applications* ISSN : 2248-9622, Vol. 3, Issue 6, Nov-Dec 2013, pp.42-46
- [6] H. -. Lou, "Implementing the Viterbi algorithm," in *IEEE Signal Processing Magazine*, vol. 12, no. 5, pp. 42-52, Sept. 1995.
- [7] Y. S. Wong, W. J. Ong, J. H. Chong, C. K. Ng and N. K. Noordin, "Implementation of convolutional encoder and Viterbi decoder using VHDL," *2009 IEEE Student Conference on Research and Development (SCoReD)*, Serdang, 2009, pp. 22-25.
- [8] A. J. Mandwale and A. O. Mulani, "Different Approaches For Implementation of Viterbi decoder on reconfigurable platform," *2015 International Conference on Pervasive Computing (ICPC)*, Pune, 2015, pp. 1-4.
- [9] J. K. Potts, "Error probabilities for Viterbi decoders with correlated noise," in *Proceedings of the Institution of Electrical Engineers*, vol. 126, no. 5, pp. 365-367, May 1979.
- [10] J. Zhang, J. You, LU and X Ma, "An iterative soft-decision decoding algorithm for conventional concatenated codes," *2009 Fourth International Conference on Communications and Networking in China*, Xian, 2009, pp. 1-5.